

Distribution Analysis of Mental Health Scores

Across Population Segmentations Using APMS 2014

NHS Adult Psychiatric Morbidity Survey 2014

CIS-R Score Distribution | Anxiety & Depression Scales | QQ Analysis

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1. Data Description:

- **APMS 2014**

The Adult Psychiatric Morbidity Survey (APMS) is a nationally representative survey of adults living in private households in England. Conducted approximately every seven years since 1993, it provides prevalence estimates for a wide range of mental health conditions.

The 2014 wave, carried out by NatCen Social Research and the University of Leicester on behalf of NHS Digital, interviewed 7,546 adults. The survey used clinical and lay interviews to assess psychiatric conditions, service use, and risk factors.

- **The CIS-R Score**

The Clinical Interview Schedule - Revised (CIS-R) is the primary continuous measure in APMS 2014. It assesses 14 neurotic symptom domains (fatigue, sleep problems, irritability, worry, depression, anxiety, obsessions, concentration, somatic symptoms, compulsions, phobias, panic, and depressive ideas) over the past week.

- **Files Used**

The sole input for this analysis is:

Primary file: apms-2014-ch-02-tabs.xls - Chapter 2: Common Mental Disorders

2. Procedure:

- **Data Loading**

Each of the nine sheets was loaded using pandas with the xlrd engine. Row and column indices for each segment were identified by inspecting the raw Excel layout. All percentage values were validated for internal consistency. The scores are grouped into 4 severity bands.

```
Sheets loaded successfully:
  cisr_age_sex      → 46 rows x 9 cols
  cisr_trend        → 45 rows x 33 cols
  cmd_age_sex       → 43 rows x 9 cols
  cmd_ethnicity     → 74 rows x 6 cols
  cmd_household     → 42 rows x 9 cols
  cmd_employment    → 72 rows x 5 cols
  cmd_benefit       → 77 rows x 7 cols
  cmd_region        → 75 rows x 10 cols
  cmd_smoking       → 72 rows x 5 cols
```

- **Distribution Reconstruction**

Since the APMS tables contain grouped percentage data (not individual-level scores), pseudo-observations were reconstructed for each population segment using the following procedure:

- For each CIS-R band, N pseudo-observations were generated proportional to the band percentage (N = 400 total per segment)
- Points were distributed uniformly within each band interval [0,6), [6,12), [12,18), [18,30)
- The resulting pseudo-sample preserves the empirical first and second moments of the grouped distribution

This approach enables standard distributional analyses (KS tests, QQ plots, ECDF comparisons) while being transparent about the grouped nature of the source data.

- **Distribution Fit Testing**

Three Types of distributions were evaluated for CIS-R score distribution:

1. Normal
2. Log- normal
3. Exponential

3.Insights:

- **Overall CIS-R Score Distribution**

The CIS-R score distribution in the English adult population is strongly right-skewed. The majority of adults (68.6%) score in Band 1 (0-5), indicating no clinically significant symptoms. The distribution follows a steep decay from the zero symptom pole, consistent with a Log-Normal or exponential-family distribution rather than a Normal distribution

CIS-R by Age & Sex (All adults):								
	16-24	25-34	35-44	45-54	55-64	65-74	75+	All
0-5	67.1	66.3	66.4	65.3	66.3	76.6	77.6	68.6
6-11	15.5	16.5	15.7	16.7	16.7	13.2	14.3	15.7
12-17	7.8	9.0	8.1	8.2	7.8	5.9	4.7	7.6
18+	9.5	8.2	9.8	9.8	9.2	4.2	3.3	8.1

- **Summary Statistics**

Weighted summary statistics computed from band midpoints (2.5, 8.5, 14.5, 22.0):

Summary statistics per group (CIS-R): [mean, std, skewness, excess kurtosis]				

Men (overall)	mean=5.28	std=5.48	skew=2.05	kurt=3.86
Women (overall)	mean=6.57	std=6.35	skew=1.51	kurt=1.49
All (overall)	mean=5.93	std=5.97	skew=1.76	kurt=2.48
Age 16-24	mean=6.22	std=6.25	skew=1.64	kurt=1.90
Age 25-34	mean=6.17	std=6.05	skew=1.63	kurt=2.02
Age 35-44	mean=6.33	std=6.32	skew=1.61	kurt=1.77
Age 45-54	mean=6.40	std=6.32	skew=1.58	kurt=1.69
Age 55-64	mean=6.23	std=6.19	skew=1.64	kurt=1.95
Age 65-74	mean=4.82	std=4.88	skew=2.22	kurt=5.17
Age 75+	mean=4.56	std=4.49	skew=2.37	kurt=6.49

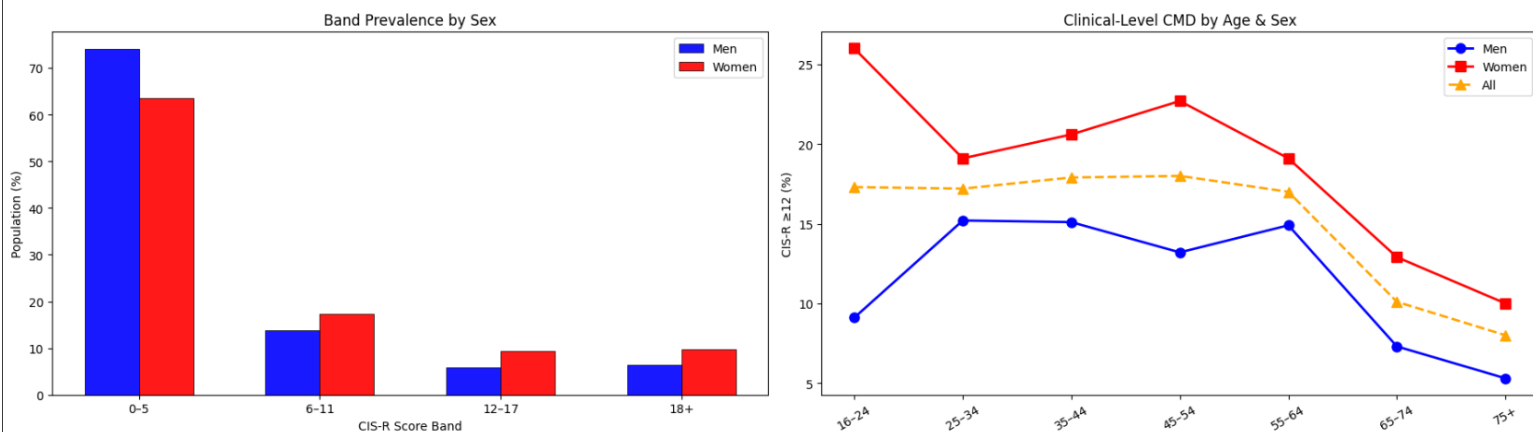
All skewness values exceed 1.5, confirming strong positive skew. The high excess kurtosis values indicate heavier tails than a Normal distribution.

• CIS-R Score Distribution by Age

Women show a significantly heavier tail at the clinically significant end of the CIS-R distribution. The key differences are as follows:

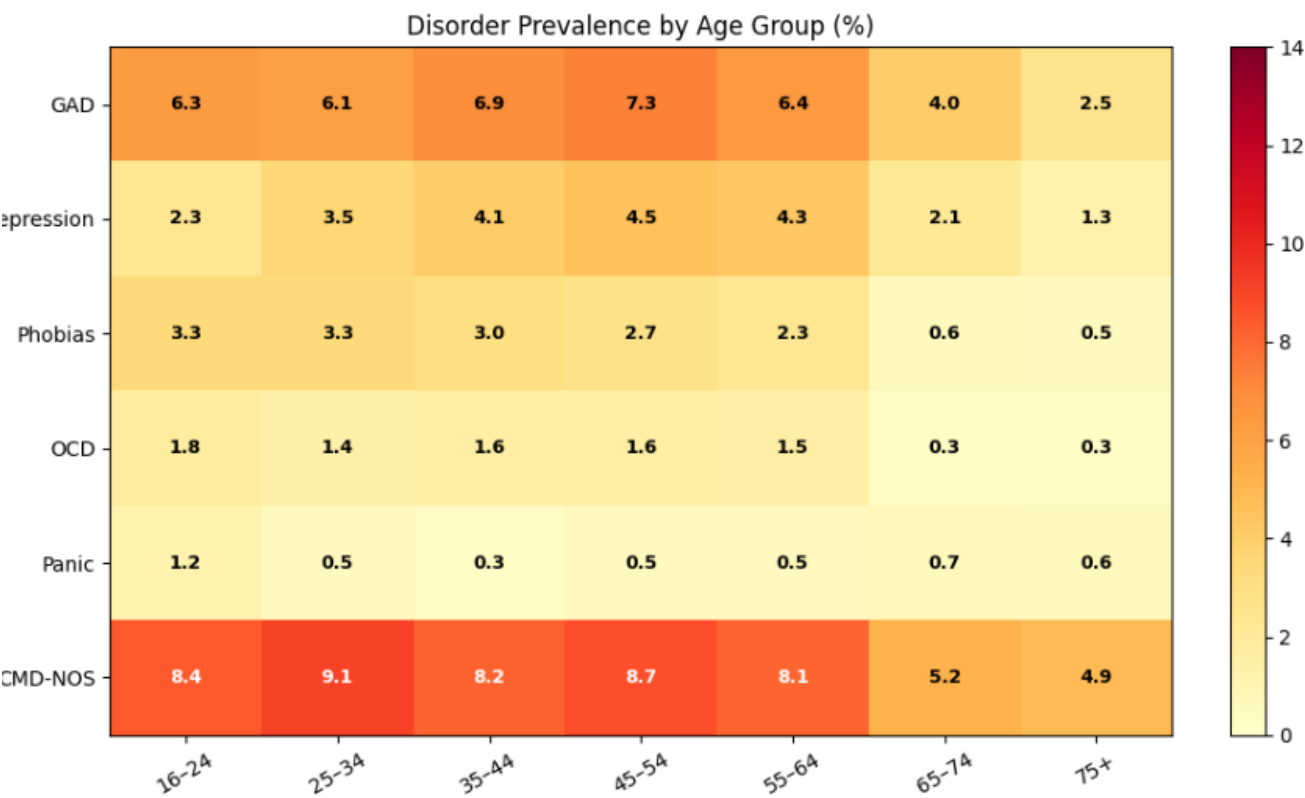
1. Women scoring CIS-R 12+ (clinical CMD): 19.1% vs Men: 12.2%
2. Women scoring CIS-R 18+ (severe CMD): 9.8% vs Men: 6.4%

CIS-R Score Distribution by Sex — APMS 2014



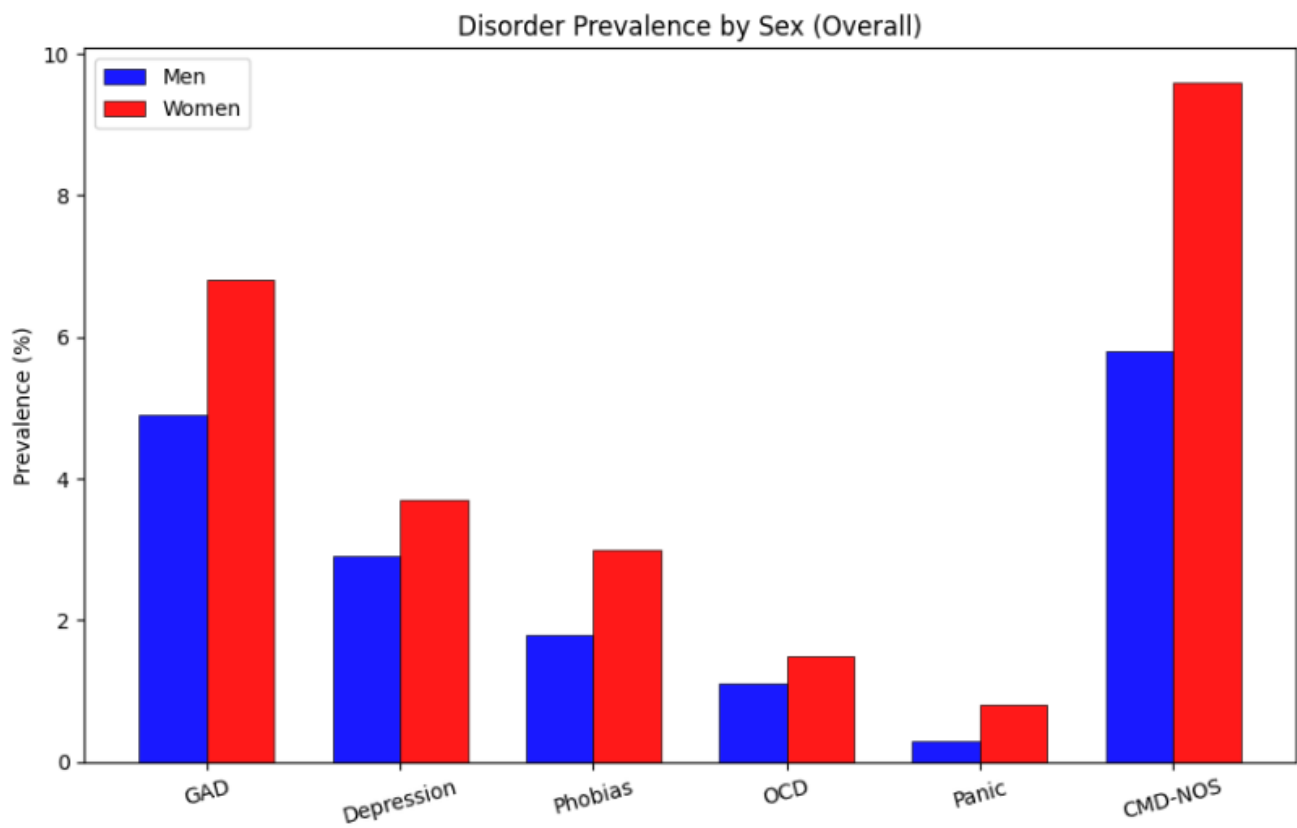
- Disorder Prevalence by Age Group

Mental health disorder prevalence generally peaks in middle-aged groups (35-54)
And declines with age, with GAD and CMD showing the highest overall rates.



- **Disorder Prevalence by Sex**

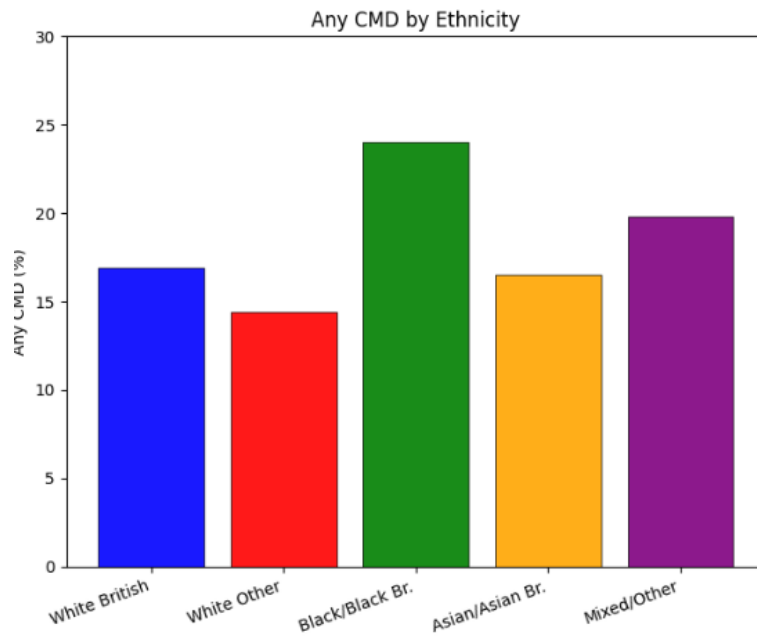
Females consistently exhibit higher disorder prevalence than males, with CMD and GAD being the most common across both groups



- **Distribution across Social Segments:**

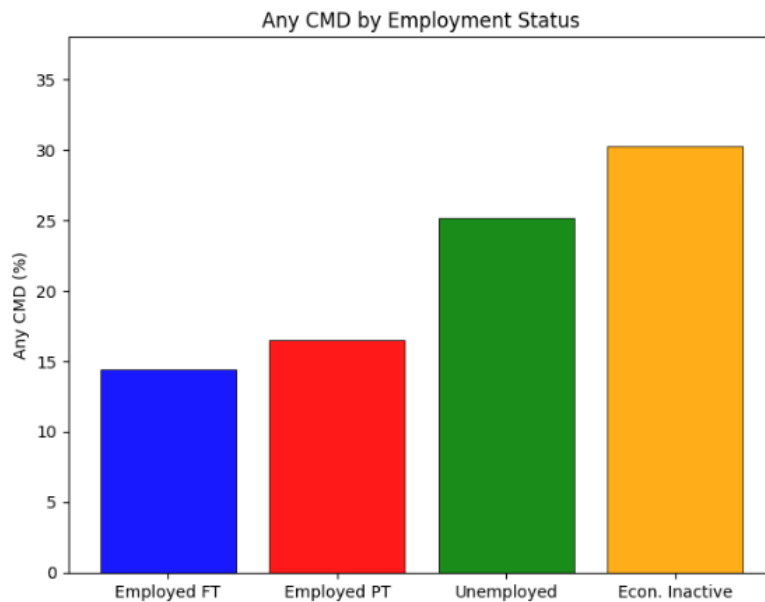
- 1. CMD Prevalence across Ethnicity:**

CMD prevalence varies across ethnicities, with the highest in Black/Black British groups and lowest on White other populations.



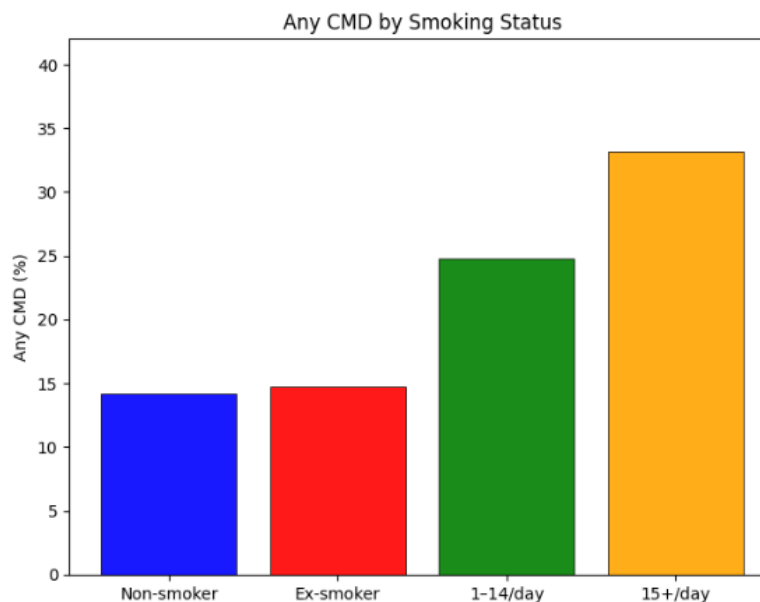
2. CMD Prevalence across Employment Status

CMD prevalence is highest among economically inactive and unemployed individuals, and lowest among in full-time employment.



3. CMD Prevalence across Smoking population

Higher smoking frequency is associated with increased CMD prevalence, with the highest rates among heavy smokers.

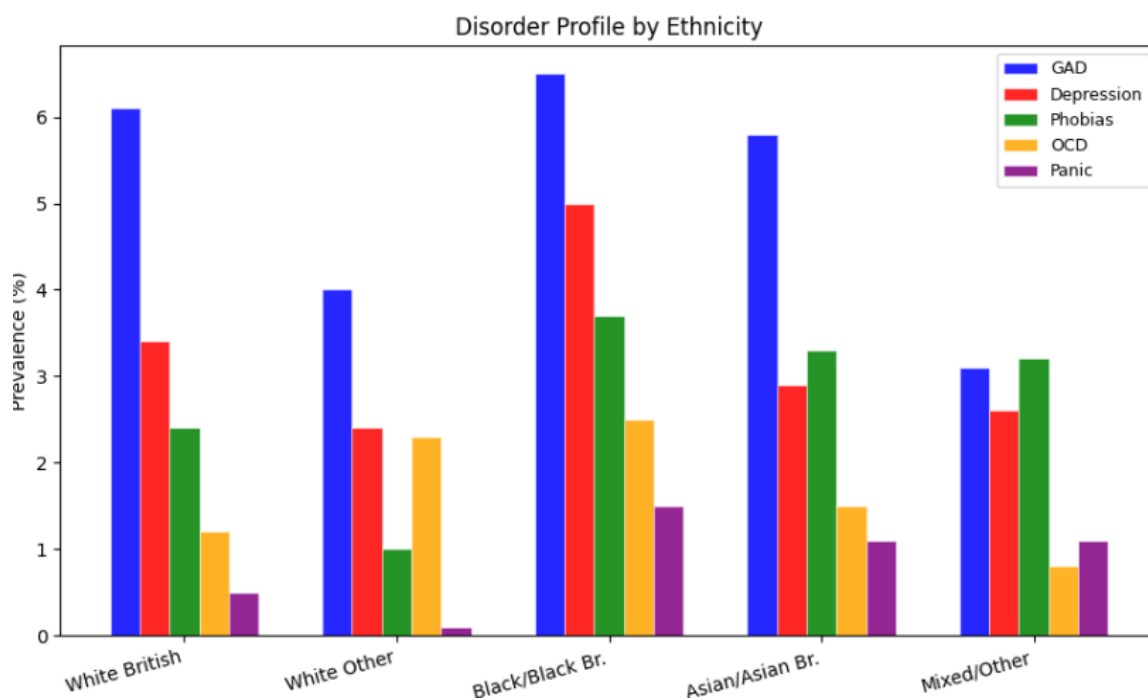


- **Disorder specific Distribution:**

- 1. Disorder Profile by Ethnicity**

The disorder-specific analysis reveals clear variation in mental health prevalence across ethnic groups and age categories.

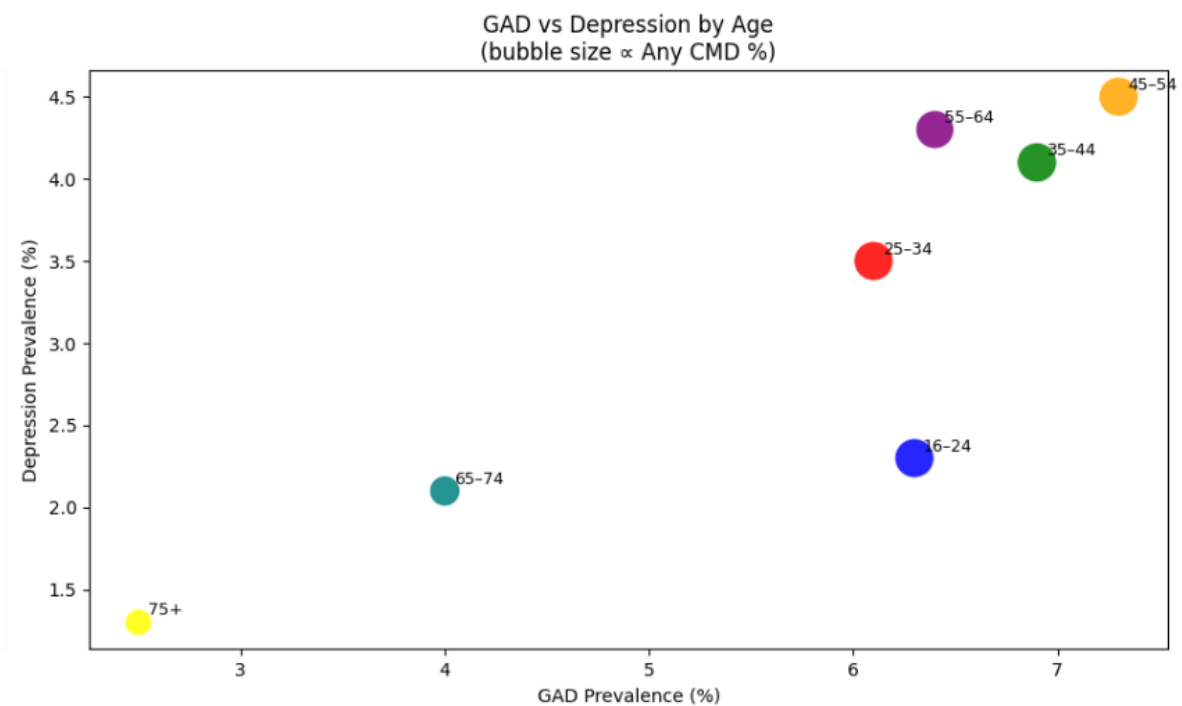
- Generalized Anxiety Disorder (GAD) is consistently the most prevalent disorder across all ethnic groups, with the highest rates observed among Black/Black British individuals.
- Depression follows a similar pattern, with elevated prevalence in Black/Black British and Asian groups compared to White groups.
- Phobias and OCD show moderate variation, while panic disorder remains relatively low across all groups.



- 2. GAD VS Depression by Age**

The bubble scatter analysis comparing GAD and depression across age groups shows:

- Peak prevalence occurs in the **35–54 age range**, where both GAD and depression are highest.
- Younger adults (16–24) show moderate levels, while older adults (65+) exhibit a clear decline.
- Bubble sizes (representing “Any CMD”) confirm that **mid-life groups carry the highest overall mental health burden**.

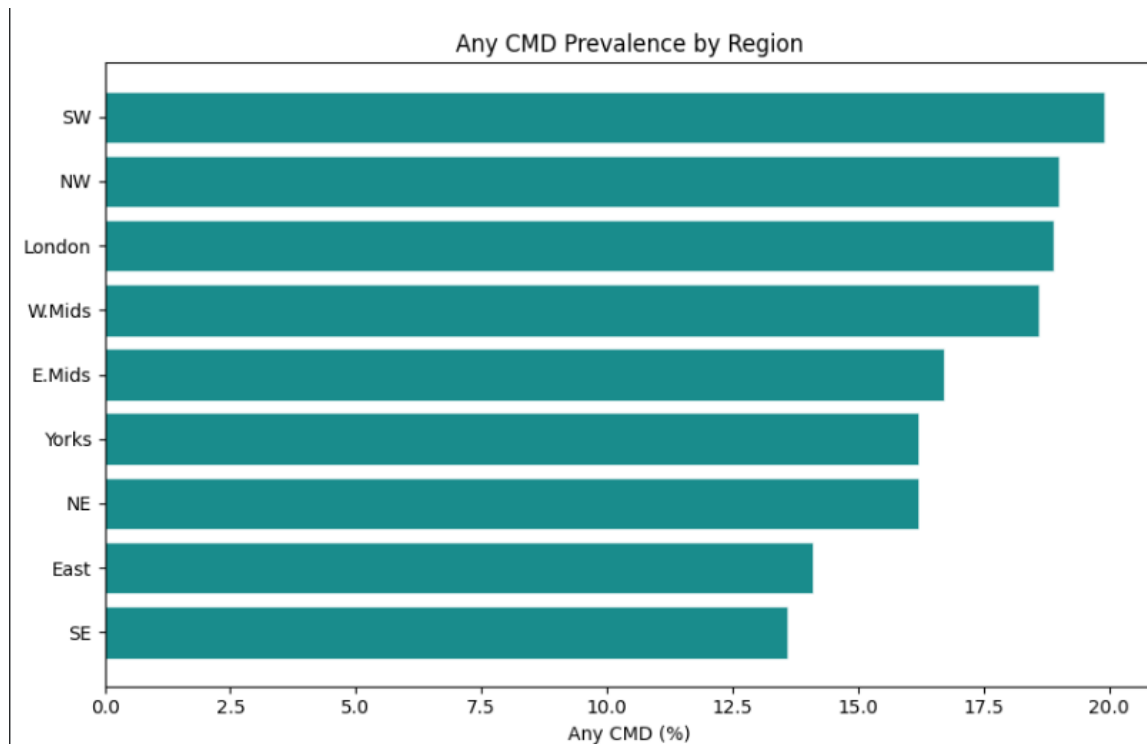


- **Regional Distribution of CMD:**

1. **Any CMD Prevalence by Region**

Regional variation in Common Mental Disorders (CMD) is present but less pronounced than socio-economic differences.

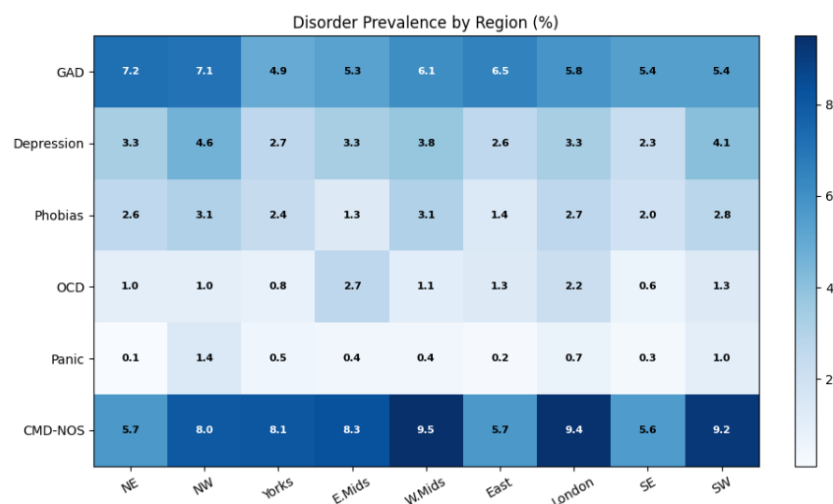
- The **South West (19.9%)**, **North West (19.0%)**, and **London (18.9%)** show the highest CMD prevalence.
- The **South East (13.6%)** has the lowest observed prevalence.



2. Disorder Prevalence by Region

The heatmap indicates:

- GAD remains the dominant disorder across all regions.
- CMD-NOS contributes significantly to total CMD prevalence.
- Regional differences are relatively modest, suggesting that **population composition (age, employment, deprivation)** may explain much of the variation.



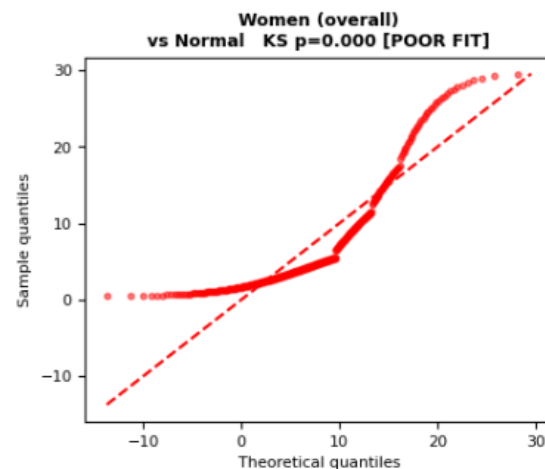
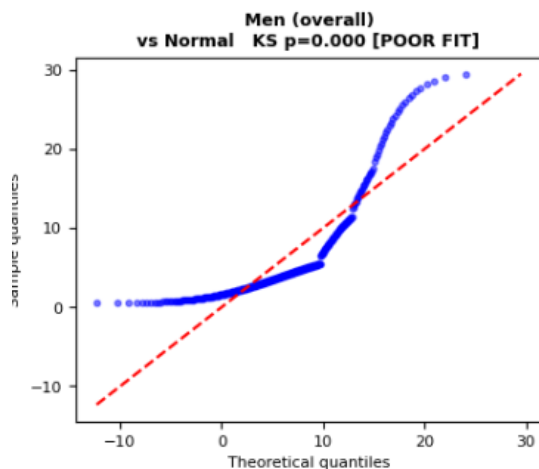
- **QQ Plots – CIS-R Score Distribution vs Theoretical Distribution:**

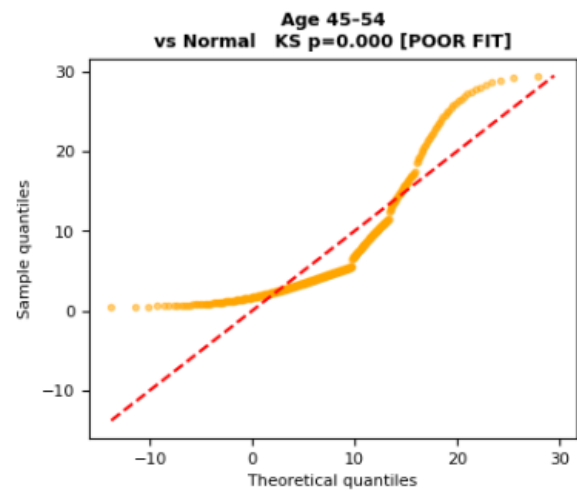
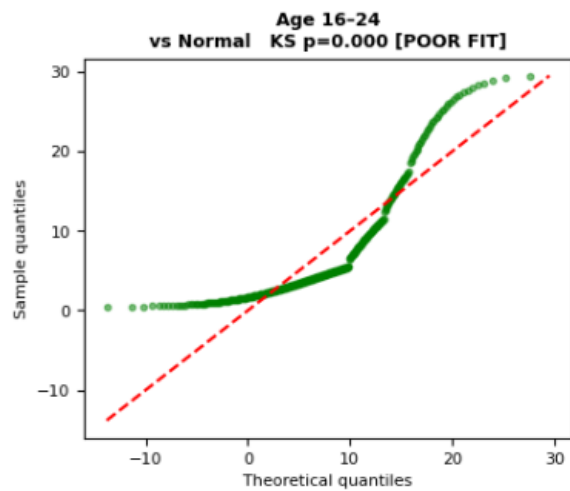
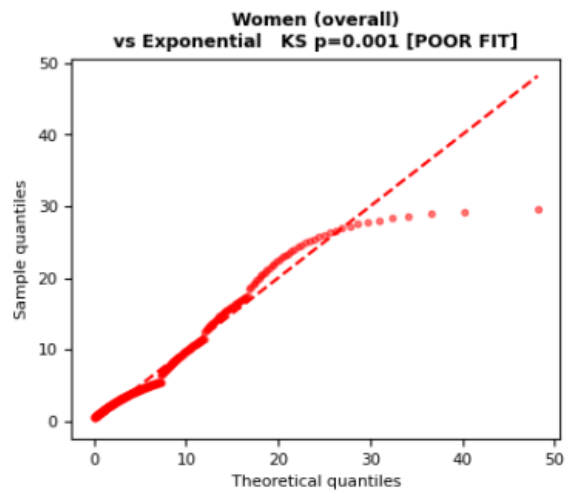
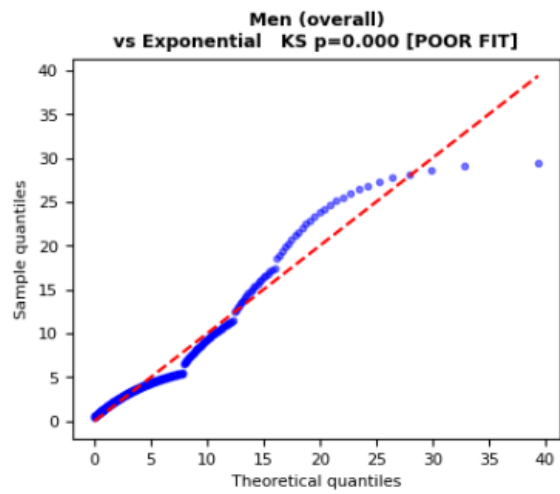
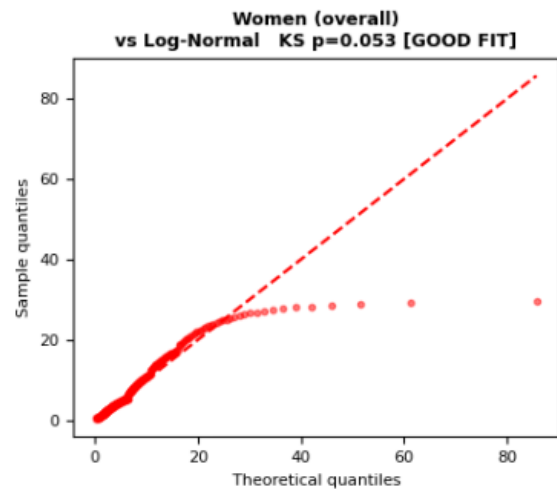
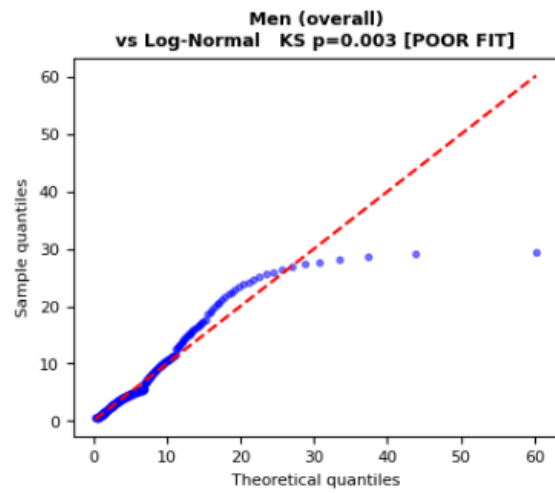
QQ plots were generated to assess the distributional properties of CIS-R scores across key population groups.

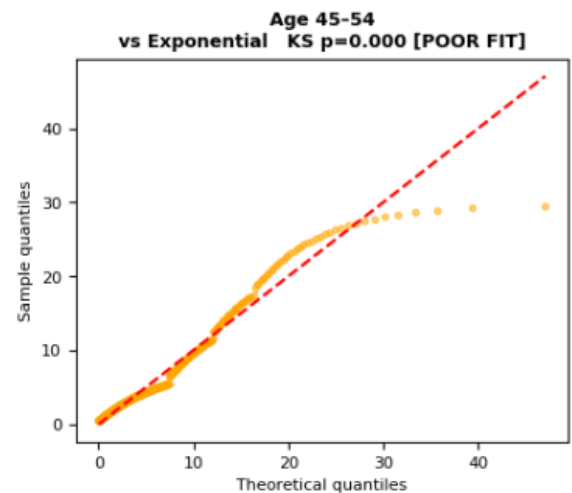
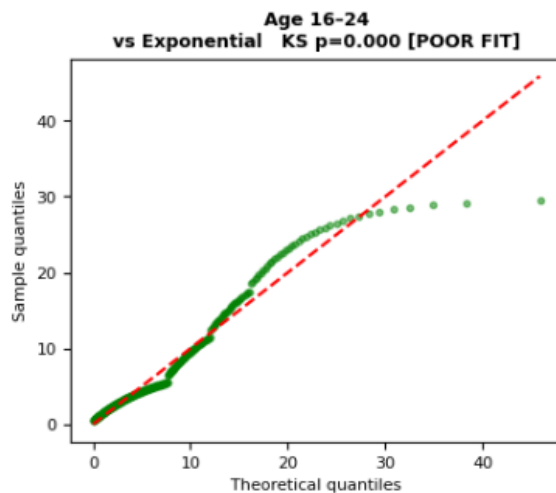
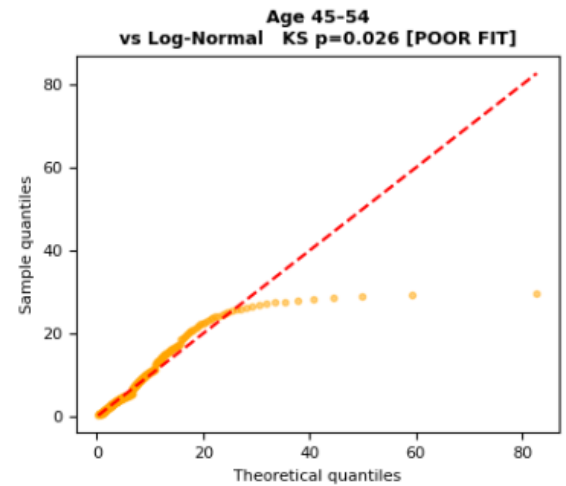
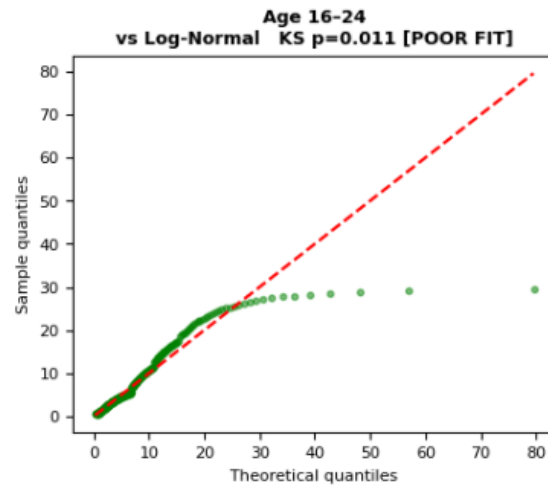
- All groups (Men, Women, Age 16–24, Age 45–54) show **strong deviation from normality**, with clear S-shaped curves.
- The Normal distribution is rejected across all cases ($p < 0.001$).
- The Log-Normal distribution provides the closest fit, particularly in the central range of the data.
- The Exponential distribution underfits the upper tail.

These findings confirm that:

- Mental health scores are **positively skewed**, with most individuals reporting low symptom levels.
- A small proportion of the population contributes disproportionately to high scores (heavy tail).







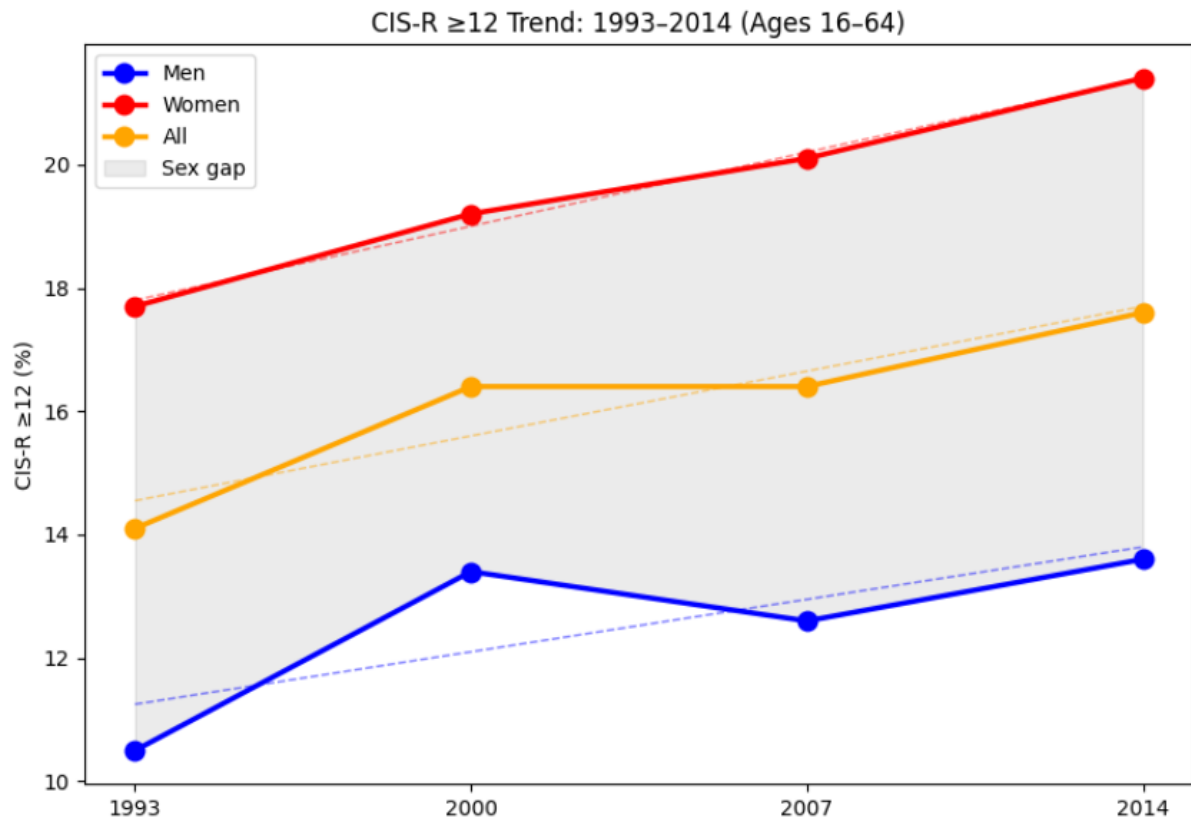
Interpretation:

- Points close to red line = good fit
- $KS\ p > 0.05$ = cannot reject that distribution
- CIS-R is heavily right-skewed $\rightarrow$ Log-Normal fits better than Normal

• Temporal Trends in CMD:

1. Temporal analysis shows a clear increase in CMD prevalence over time.

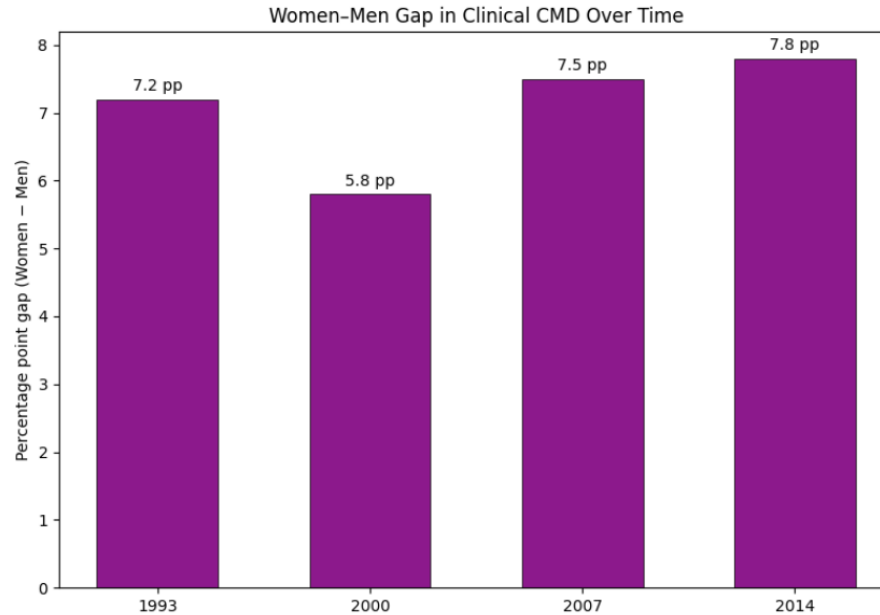
- Women: increased from **17.7% (1993)** to **21.4% 2014**
- Men: increased from **10.5% to 13.6%**



2. The gender gap:

- Narrowed slightly in 2000 but widened again to **7.8 percentage points in 2014**

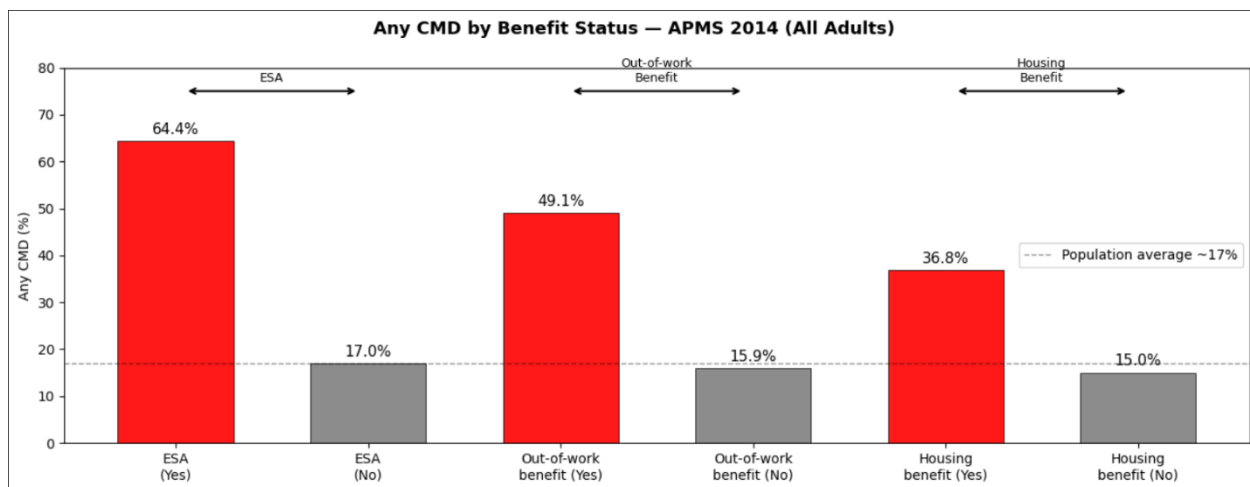
- Indicates persistent and growing inequality in mental health burden



• **CMD by Benefit Status:**

Benefit status shows the strongest association with CMD prevalence.

- ESA claimants: **64.4% CMD prevalence**
- Out-of-work benefit recipients: **49.1%**
- Housing benefit recipients: **36.8%**
- General population baseline: ~17%



This represents:

- A **3–4x increase** in CMD prevalence among vulnerable groups

- Strong evidence of **socio-economic gradients in mental health**
- **Severity Gradient Across Population Segments:**

High Risk (>20% CMD)

- Heavy smokers (15+ per day): **33.2%**
- Economically inactive: **30.3%**
- Unemployed: **25.2%**
- Black/Black British: **24.0%**

Moderate Risk (14–20%)

- Women: **19.1%**
- Age 45–54: **18.0%**
- Age 35–44: **17.9%**

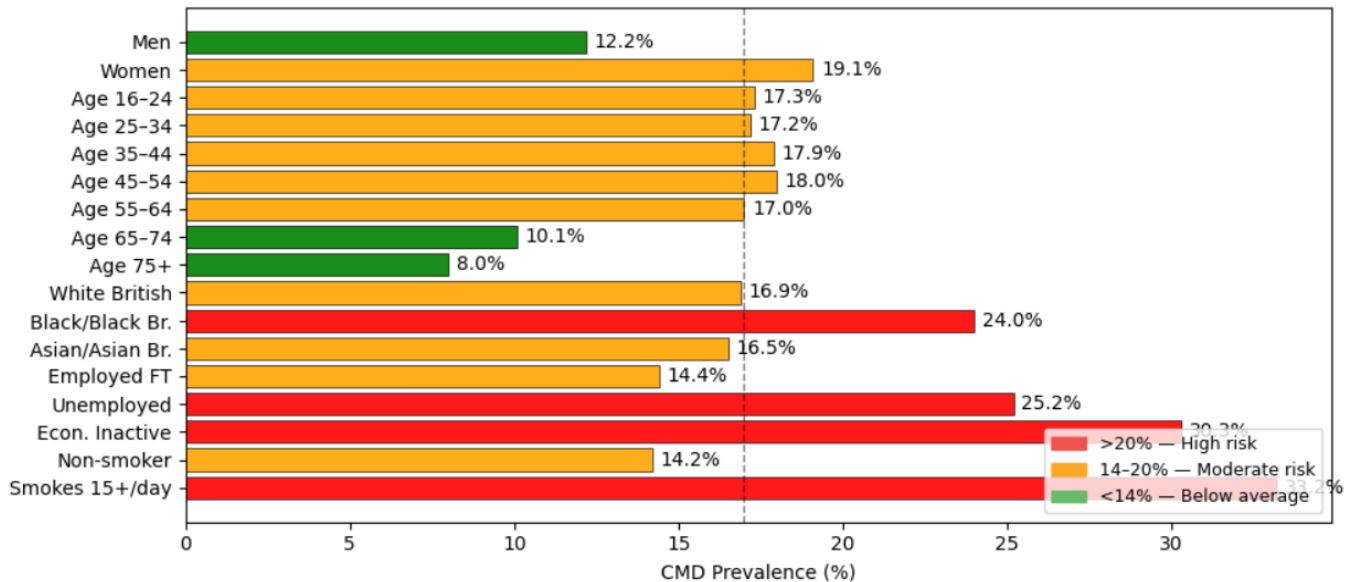
Lower Risk (<14%)

- Men: **12.2%**
- Age 75+: **8.0%**

This confirms that:

- **Socio-economic status is the strongest determinant**
- Demographic factors (age, gender) are secondary but still important

CMD Prevalence Across All Segments — APMS 2014
(CIS-R ≥ 12 for sex/age; Any CMD % for others)



4. Key Findings:

1. DISTRIBUTION SHAPE:

- CIS-R is strongly right-skewed (most people score 0–5).
- Log-Normal fits better than Normal (QQ plots, KS test).

2. SEX DIFFERENCES:

- Men CIS-R ≥ 12 : 12.2%.
- Women CIS-R ≥ 12 : 19.1%.
- Gap = 6.9 pp.

3. AGE PATTERN:

- Age 16–24: 17.3% score CIS-R ≥ 12 .
- Age 45–54: 18.0% score CIS-R ≥ 12 .
- Age 75+: 8.0% score CIS-R ≥ 12 .

4. HIGHEST RISK SEGMENTS (Any CMD):

- Smokes 15+/day: 33.2%.
- Econ. Inactive: 30.3%.
- Unemployed: 25.2%.
- Black/Black Brown: 24.0%.
- Women: 19.1%.

5. DISORDER BREAKDOWN (All adults):

- **GAD: 5.9%.**
- **Depression: 3.3%.**
- **Phobias: 2.4%.**
- **OCD: 1.3%.**
- **Panic: 0.6%.**
- **CMD-NOS: 7.8%.**

6. TEMPORAL TREND:

- **Women: 17.7% (1993) → 21.4% (2014) [+3.7 pp].**
- **Men: 10.5% (1993) → 13.6% (2014) [+3.1 pp].**
- **Sex gap has widened over 20 years.**